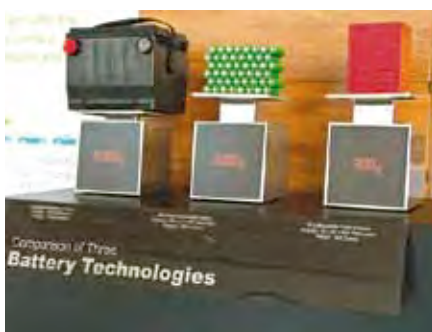




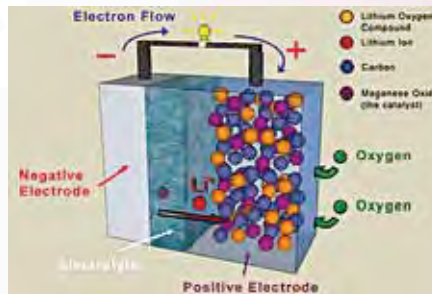
delay, thus resulting in lesser speeds. Solutions range from altering transistor material to using optical components for integrated circuits, both currently expensive methods. Intel has been working on a new 3D integrated circuit, wherein the components are arranged both vertically and horizontally in a single circuit. This allows for more transistors within a smaller volume of space, thus following Moore's Law and providing increased performance. This would result in shorter interconnects, significant reduction in power consumption, lesser production costs, and a brand new range of design possibilities.

Battery

The battery is the quintessential component of just about every mobile device. They work by converting chemical reactions into electrical energy, when electrons flow from a positively charged cathode to a negatively charged cathode, passing through an electrolyte substance. This difference in charges thus produces an electrical current. Power management in desktop systems is important, but ever more so for smart phones and tablets. Their mobility and performance is defined by just how long you can go off a single charge. However, the technology itself has remained unchanged for more than 8 years. This is evidenced by no less than the newest iPad, whose battery takes up about 90% of its total weight. Lithium Ion



IBM showcases a comparison between current battery technologies and it's upcoming Lithium-Air Cells



This diagram showcases the movement of electrons between electrodes within a lithium-air cell

cells show good energy density, with very little discharge when not in use. They rose to popularity because of their recharge ability, thanks in due to the lithium compound present as the electrode material.

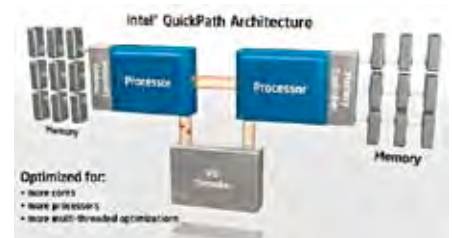
There are drawbacks: Lithium ion cells also have the disadvantage of gradually losing capacity as they're used for a longer time. There are also dangers of these cells facing leakage, or worse, combustion, if they are over-charged. The sheer number of safety measures incorporated to prevent this from happening take up valuable space in the battery, thus making it bulkier.

But researchers have been looking for alternatives. Some propose moving on from lithium to other metal compounds. IBM is hopeful of its use of air. The cathodes are made of carbon rather than lithium, which then react with oxygen in air to produce electricity. However, there are issues with getting the battery to recharge. Plus, lithium ignites when in direct contact with moist air. Currently solutions are being worked on to rid the battery of this issue, and we could see developments by 2013, with plans for commercialization by 2020.

Front-Side Bus

The Front-Side Bus or FSB is a little-known but very vital component for every system. As part of the motherboard, it is the central link between the CPU and other devices within a computer. The overall bandwidth that

FSB can provide is dependent on its clock frequency, its data path width and the number of data transfers it performs per cycle. The problem arises with the number of clock cycles an FSB can perform. If it goes by the frequency set by the motherboard, which leads to a set number of clock cycles, then overall bandwidth will be limited. Hence no matter how much faster CPUs get, their speed will always be constrained if the FSB can't keep up, as the CPU remains idle for one or more clock cycles until the memory returns its value. Memory in a system is also accessed via the front-side bus, which limits overall transfer speeds since bandwidth is being utilized for this purpose. Both AMD and Intel are working on their own advancements over the old FSB, HyperTransport and



Intel showcases it's new QuickPath Interconnect technology to replace front-side buses in the future

Quickpath Interconnect respectively, which promise to provide point-to-point connections over the use of the FSB as a central connecting point. Using this AMD's HyperTransport, for instance, allows for a theoretical transfer rate of 51.2 GB/s of aggregated throughput. To ensure none of the bandwidth for these point-to-point links is wasted, a separate memory controller on the CPU itself is used to access RAM. Currently, AMD and Intel employ their respective technologies within various components, and they are being tested for replacing front side bus in motherboards. **d**

You can read the rest of this article (which covers all 10 tech bottlenecks): http://dgit.in/cover_bottleneck